

Problem 1: Entropy

Background

Shannon entropy is defined as

$$H(X) = - \sum_{x \in X} p(x) \log p(x) \quad (\text{with the convention } 0 \log 0 := 0).$$

Problem 1A (Discrete case)

On each repeated trial of the same stimulus condition, a neuron produces a discrete response

$$X \in \{S, P, B\}$$

where S = silent, P = spike, B = burst. Let $p(X) = (p_S, p_P, p_B)$ be the probability mass function (PMF), with $p_S, p_P, p_B \geq 0$ and $p_S + p_P + p_B = 1$.

Now suppose recording tells you that the neuron is silent with probability

$$p_S = s \quad \text{for some fixed } 0 < s < 1,$$

but you are free to choose how the remaining probability $1 - s$ is split between P and B . Among all PMFs satisfying $p_S = s$, find:

1. the PMF(s) that *minimize* $H(X)$ and the minimum value,
2. the PMF that *maximizes* $H(X)$ and the maximum value.

Problem 1B (Continuous bounded case)

On each repeated trial of the same stimulus condition, suppose the neuron's response is a *continuous* firing rate

$$X \in [a, b], \quad \text{with } a < b,$$

and let $f(x)$ denote the probability density function (PDF) of X on $[a, b]$:

$$f(x) \geq 0, \quad \int_a^b f(x) dx = 1.$$

The (differential) Shannon entropy of X is defined by

$$h(X) = - \int_a^b f(x) \log f(x) dx,$$

where $\log$ denotes the natural logarithm. Among all PDFs f supported on $[a, b]$ (i.e., satisfying $f \geq 0$ and $\int_a^b f(x) dx = 1$), find the PDF that *maximizes* $h(X)$ and compute the maximum value of $h(X)$.